

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-06

BOTANY

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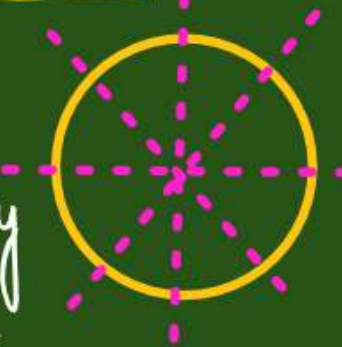
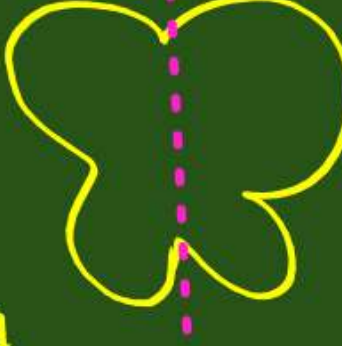


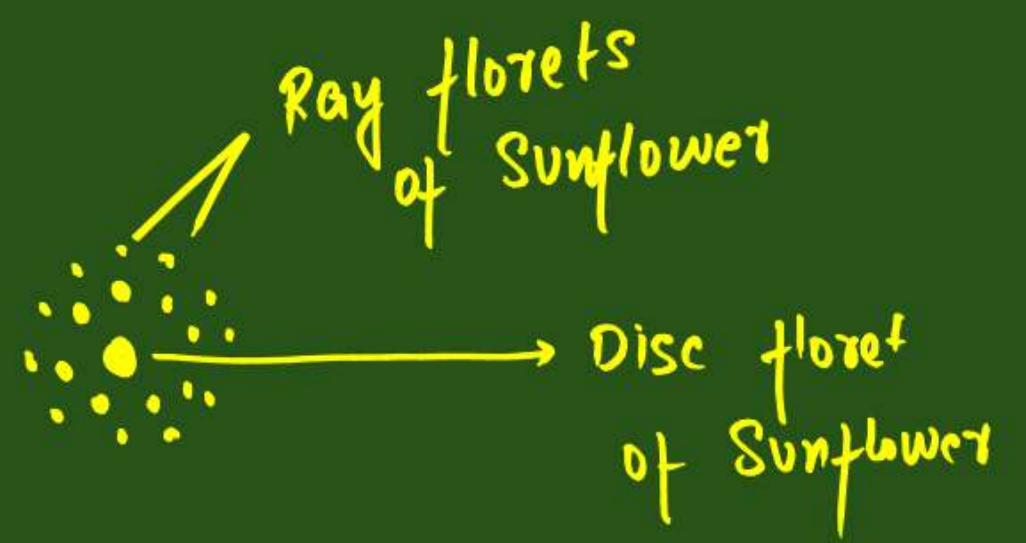
Topics To Covered

1 Structure of flower(Contd...)

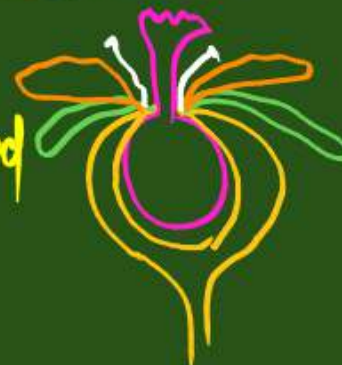
2 ...

FLOWER TYPES (ON THE BASIS OF SYMMETRY)

ACTINOMORPHIC ⊕	ZYGOMORPHIC %	ASYMMETRIC \$
Ex. <u>Mustard</u>, <u>Datura</u>, <u>Chilli</u>, <u>Disc floret of sunflower</u> → Radial Symmetry → A flower can be divided into two equal halves by any plane passing through centre 	Ex. <u>Pea</u>, <u>Bean</u>, <u>Cassia</u>, <u>Gulmohar</u>, <u>Ray florets of sunflower</u> → Bilateral Symmetry → A flower which can be divided into two equal halves by only one plane passing through centre. 	Ex. <u>Canna</u> → No Symmetry (Asymmetrical flower)



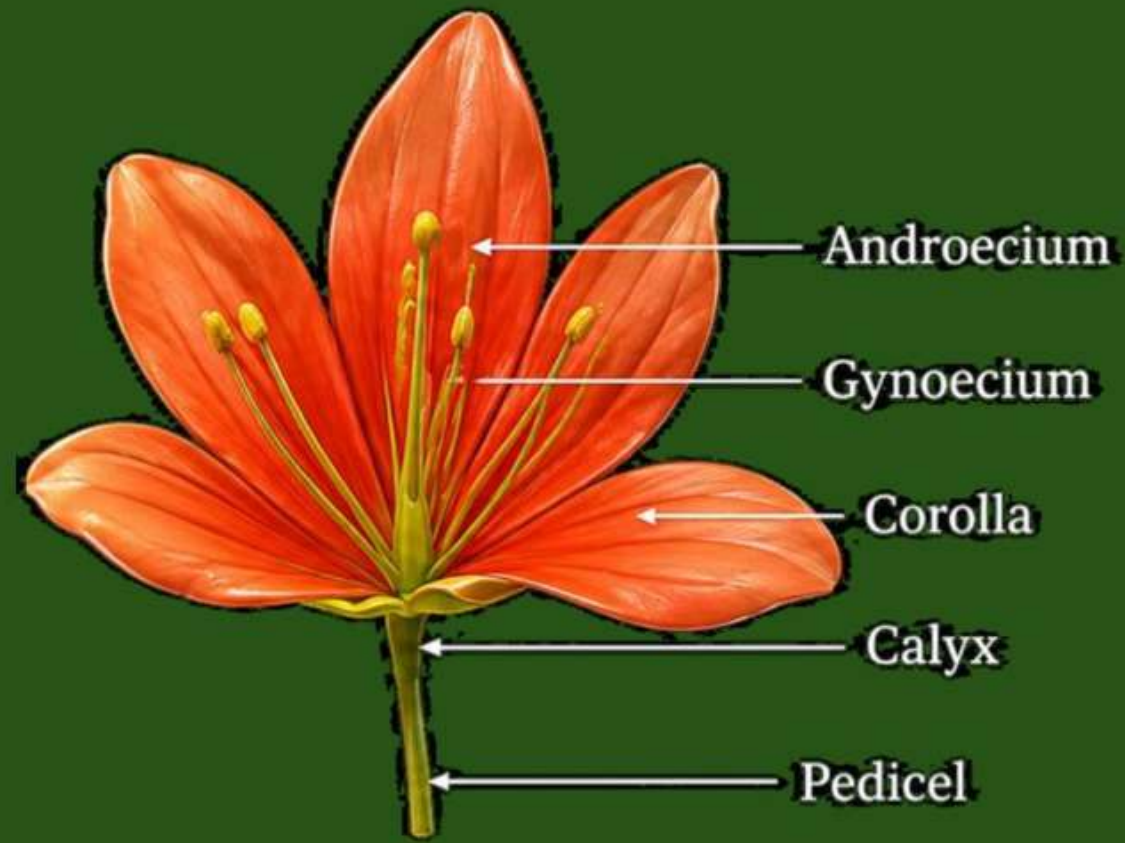
FLOWER TYPES (ON THE BASIS OF OVARY POSITION)

HYPOGYNOUS	PERIGYNOUS	EPIGYNOUS
Ex. <u>Mustard</u>, <u>China rose</u>, <u>Brinjal</u> → Here ovary occupies the top most position and other floral parts arise from base of ovary. → Superior ovary { $\overline{G}_1$ } 	Ex. <u>Plum</u>, <u>Rose</u>, <u>Peach</u> → ovary is half filled inside thalamus → ovary is half inferior or half superior } "$\overline{G}_1$" 	Ex. <u>Guava</u>, <u>Cucumber</u>, <u>Ray florets of sunflower</u> → ovary is completely embedded inside thalamus → ovary is inferior ($\underline{G}_1$) 

PARTS OF FLOWER

04 whorls of flower:

- Calyx
- Corolla
- Androecium
- Gynoecium



Calyx



Corolla



Androecium Gynoecium

PARTS OF FLOWER

1. Calyx: (K)

- Outermost whorl.
- It is group of Sepals
- It is generally green in colour.
- Performs photosynthesis.
- It protects the flower during Bud Stage.

→ Fused Sepals = Gamosepalous - $K(n)$
 $n = \text{no. of Sepals}$

→ Free Sepals = Polysepalous - Kn

2. Corolla: "C"

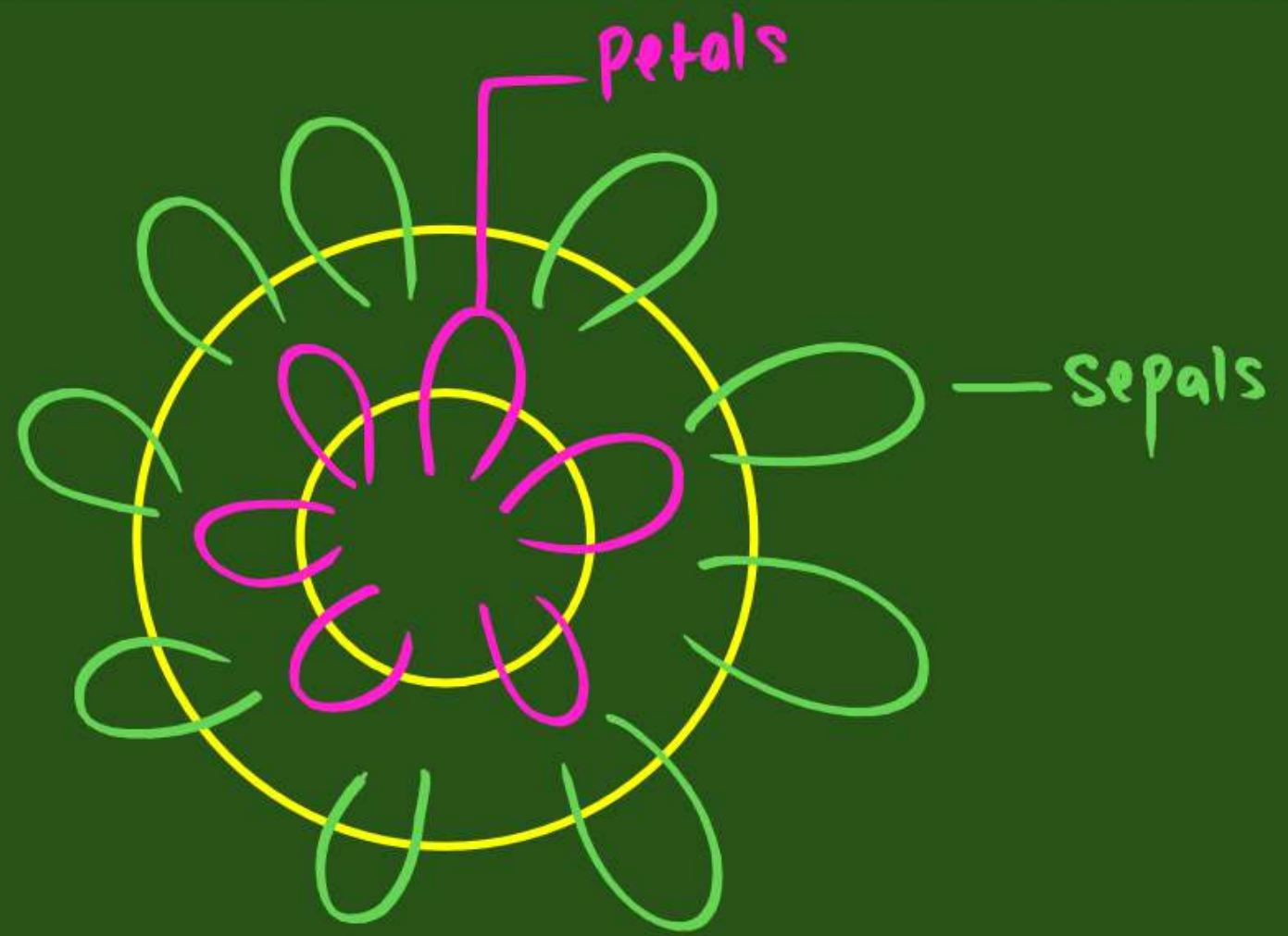
- present inner to Calyx.
- It is group of petals.
- Petals are coloured and attracts pollinators for pollination.

→ Corolla - different shapes

- Funnel Shaped
- wheel Shaped
- Tubular Shaped
- Cross Shaped

→ Fused petals = Gamopetalous - $C(n)$

→ Free petals = Polypetalous - Cn



AESTIVATION

→ Pattern of arrangement of Sepals/Petals of Same Whorl.
 → 04 types:



→ AKA Papilionaceous aestivation.

→ Seen only in Petals

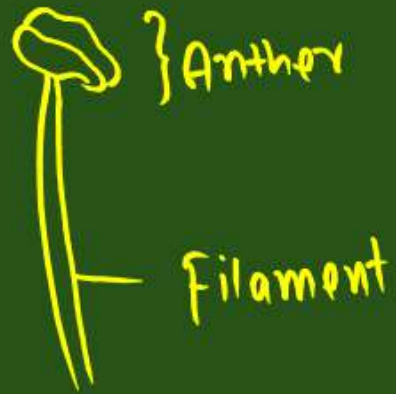
VALVATE	TWISTED	IMBRICATE	VEXILLARY
<u>Calotropis</u>  → Margins of Sepals/petals are just touching each other but does not overlap.	<u>China Rose, Lady's finger, Cotton</u>  → One margin of each sepal/petal is placed inside and second margin towards outside	<u>Cassia, Gulmohar</u>  → One sepal/petal has both margins inside. → 2nd sepal/petal has both margins outside → Rest all the sepals/petals are just like twisted aestivation.	<u>Pea, Bean</u>  → Vexillum/Standard • 01 in number • Posterior end. → Lateral wings • 02 in no. → Keel • 2 in no. • Fused • Anterior end

largest petal

ANDROECIUM

"A"

- Male reproductive whorl.
- It is group of Stamens.
- Stamen — two parts
 - Filament → Stalk like Str. → Sterile.
 - Anther → Fertile part. → produces pollen grains (♂ gametophyte).



Note: 1. Sterile Stamen is called as Staminate.
2. No. and length of Stamens is different in different species. Ex. Salvinia, Mustard.

ADHESION

OF STAMEN

↓
NEXT Lecture.

COHESION OF STAMEN

↓
NEXT lecture.

GYNOECIUM

Next lecture.



PLACENTATION

Next lecture.



MARGINAL	AXILE	PARIETAL	FREE CENTRAL	BASAL
Pea 	China Rose, Tomato, Lemon 	Mustard, Argemone 	Dianthus, Primrose, Chilly 	Sunflower, Marigold 

QUESTIONS

In a pea flower, five petals are arranged in a specialized manner with one of posterior, two lateral and two anterior. These are named as, and respectively. [NEET-2023]

- A** Keel, Wings and Standard
- B** Vexillum, Keel and Standard
- C** Keel, Standard and Carina
- D** Standard, Wings and Keel

Thank
You